Tobias Mwangi-Hartley

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PROFESSIONAL SUMMARY

Data scientist with experience developing forecasting, churn propensity and experimentation solutions using Python, SQL and R. Hands-on with feature engineering, model evaluation, scheduled notebook workflows, batch scoring and automated tests. Now seeking to grow into machine learning engineering, building on applied modelling experience and practical collaboration with data platform teams.

EXPERIENCE

Data Scientist | Thistlebox *2022–Present*

- Work across experimentation, forecasting and churn modelling.
- In 2025, began writing pytest unit tests for forecasting feature code; coverage is approximately 60%.
- Delivered internal training, “Stats for PMs,” three times.

Junior Data Analyst | Firth Energy Retail *2020–2022*

- Segmented customers and analysed tariffs using R and SQL.
- Automated monthly regulator reports with R Markdown, saving the team about two days each month.

SELECTED PROJECTS

Recipe Demand Forecasting | Thistlebox *2022–2024*

- Built weekly forecasts of orders per recipe and region, six weeks ahead, using LightGBM and calendar, price and menu-position features.
- Ran the workflow as a scheduled notebook job in Vertex AI Workbench, writing results to BigQuery for operations to use in Looker.
- Improved WAPE from 23% to 14%; estimated food waste reduction was 9%.
- Technologies: Python, LightGBM, BigQuery, Vertex AI Workbench, Looker.

Churn Propensity and Retention Targeting | Thistlebox *2024–2025*

- Built an XGBoost churn model scored weekly in BigQuery; retention offers were sent to top-decile customers.
- In a 10% holdout validation, churn decreased by 2.1 percentage points in the treated group.
- Technologies: Python, XGBoost, SHAP, BigQuery, dbt.

Sequential A/B Testing Calculator | Thistlebox *2023*

- Built a Streamlit app for planning and analysing experiments with sequential boundaries, used by approximately 25 people in product and growth.
- Wrote the app and its Dockerfile; the data platform team handled deployment to Cloud Run.
- Technologies: Python, Streamlit, statsmodels, Docker.

Image Classification Course Project | Personal *2025*

- Fine-tuned a small CNN on a public food images dataset as part of an online deep learning course; achieved approximately 80% accuracy. Learning project only.
- Technologies: PyTorch, torchvision, Google Colab.

EDUCATION

MSc in Statistics
Dissertation: Bayesian hierarchical models.

SKILLS

Programming and analysis: Python, pandas, scikit-learn, R, SQL
Machine learning and statistics: LightGBM, XGBoost, PyTorch, feature engineering, forecasting, churn propensity modelling, SHAP, experimental design, Bayesian hierarchical models
Data and workflow tools: BigQuery, Vertex AI Workbench, Looker, dbt, R Markdown
Applications and testing: Streamlit, Docker, pytest

ADDITIONAL INFORMATION

UK settled status